Group Project Instructions

1. Purpose and Focus

The goal of this group project is to bring together everything you have learned in this course and apply it to a complete entry-level data analysis project. You will work with a real dataset to:

- clean and prepare data (data wrangling)
- compute descriptive statistical measures (descriptive analytics)
- create effective data visualizations (descriptive analytics)
- conduct simple and multiple linear regression (predictive analytics)
- answer business-relevant research questions (business insight)
- communicate meaningful insights using a well-structured Jupyter Notebook (business communication)

1.1. A Semi-Open Project

This project is semi-open, meaning:

1) It is not as fixed and specific as homework or the midterm.
2) It is not fully open-ended either.
3) You are given:
   - a dataset
   - a business context
   - expectations for structural data analysis
   - several guiding research questions (required)
4) You are also encouraged to explore one additional question of your own interest at the end.

1.2. Group Formation

Groups will be formed randomly in class. **4 students per group**. All group members receive the same grade unless there is evidence of non-participation or free rider.

2. Deliverables and Requirement

2.1. Deliverable

Each group must submit one Jupyter Notebook (.ipynb). This notebook will also serve as your presentation file in **Week 15** (presentation on Dec 11, no class/Q&A on Dec 9), so no separate slides/PPT are required. Each group will have **20 minutes for presentation + 5 minutes for Q&A**. Please follow the required notebook structure

outlined below.

2.2 Required Structure of the Jupyter Notebook

1) *Title and Introduction*
   - Dataset name
   - Business context
   - Overview of research questions (both required + your own)

2) *Setup*
   - Import necessary libraries (pandas, numpy, matplotlib, statsmodels/sklearn)
   - Load the dataset
   - Show first few rows & basic info

3) *Data Wrangling / Cleaning*

You must demonstrate the following:

- Checking and handling duplicates
- Checking and handling missing values (drop or impute, explain your choice)
- Detecting outliers (boxplots, z-scores, IQR) and deciding how to handle them
- Correcting data types if needed
- Creating at least 1 – 2 new variables if appropriate (e.g., total revenue, CTR = clicks/impressions)

Document your work clearly in Markdown.

4) *Descriptive Analysis*

Using appropriate statistical measures to understand the key variables based on those guiding research questions (NO need to analyze every variable):

- Mean, median, percentiles, variance, skewness, correlation matrix

Explain what each descriptive result means for the business context.

5) *Data Visualization*

Visualize only the key variables or relationship you care about based on those guiding research questions (DO NOT plot everything, which makes the notebook cluttered). Examples of figures: histogram, boxplot, line plot or bar chart, scatterplot, etc.

Each figure must include:

- A clear title and labeled axes
- A short 1–2 sentence interpretation below the plot

6) *Linear Regression Analysis*

You must run:

- At least one simple linear regression
- At least one multiple linear regression

Requirements:

- Identify dependent variable (Y) and predictors (X)
- Explain why these variables make sense for the business context
- Interpret signs, magnitudes, and significance of regression coefficient
- Comment on R^2 or adjusted R^2
- Present business implications

Only use regression techniques covered in class. Models or methods we did NOT cover will not be credited.

7) Answering the Required Research Questions

Each dataset will have:

- 3 – 4 required guiding questions
- You must answer them based on the above descriptive (statistical measure, data visualization) and predictive analysis (regression analysis)

8) One Open-Ended Question (Your Choice)

You must pose and answer one question of your own interest based on proper data analysis, such as:

- Which customer segment is most profitable?
- What drives price variation?
- Which employees are at highest attrition risk?
- What hours have the highest demand?

9) Conclusion / Business Insights

Summarize:

- What you learned
- Key insights
- Recommendations for the business
- Limitations of the analysis

10) Code Quality & Notebook Organization

Your notebook should demonstrate:

- Clean, readable code
- Proper use of markdown to explain results
- Logical structure
- High-quality plots
- No unnecessary packages or unused code (unless necessary)
- No complex ML/AI packages not taught in class

3. Dataset Options and Guiding Questions

Each group will be randomly assigned one of the following three datasets. For your assigned dataset, you will:

- download the dataset from Kaggle,
- review examples of shared notebooks on Kaggle for inspiration (do not just copy),
- conduct data cleaning, descriptive analysis, visualization, and regression,
- answer the required guiding questions,
- explore **one additional open-ended question of your own interest**.

Dataset 1: Hotel Booking Demand

This dataset contains booking records from a city hotel and a resort hotel, including information on guest characteristics, booking channels, stay details, and pricing.

The business objective is to understand hotel pricing dynamics, guest behavior patterns, and key drivers of revenue to support revenue management and operational decisions.

You can download this dataset "*hotel_bookings.csv*" in this Kaggle page, which provides a detailed data dictionary, variable descriptions under the **Data Card** tab, and hundreds of shared example notebooks under the **Code** tab, which you may use for inspiration (*do not copy*).

Required Guiding Questions (Answer all)

1. (Descriptive Analysis) What are the key patterns in guest behavior (e.g., number of adults/children, customer type, stay length)? Provide descriptive statistics and at least one visualization to summarize these patterns.

2. (Exploratory Analysis) How does ADR (Average Daily Rate) differ across major categories, such as: hotel type (city vs. resort), customer type (transient, group, contract), meal type or market segment. Visualize and interpret whether these groups show meaningful differences.

3. (Focused Exploration) Is there a relationship between lead time and ADR? (Do bookings made far in advance pay more, less, or about the same?). Interpret the observed relationship and discuss possible business reasons behind it.

4. (Regression Analysis) Build a multiple linear regression model predicting ADR using 2 – 4 relevant predictors (e.g., hotel type, guest count, lead time, stay duration, market segment). Interpret the coefficients clearly in business terms, explaining how each variable affects hotel pricing while holding other factors constant.

Dataset 2: HR Analytics

This dataset contains employee information such as job role, education level, years of experience, industry, company size, and salary/income.

The business objective is to understand what drives salary differences, evaluate whether compensation is structured fairly, and provide insights for talent management and HR strategy.

You can download this dataset "*WA_Fn-UseC_-HR-Employee-Attrition.csv*" in this Kaggle page, which provides a detailed data dictionary, variable descriptions under the **Data Card** tab, and hundreds of shared example notebooks under the **Code** tab, which you may use for inspiration (*do not copy*)..

Required Guiding Questions (Answer all)

1. (Descriptive Analysis) What are the important patterns in employee demographics and job attributes (e.g., experience years, education, job title, industry, company size)? Provide descriptive statistics and at least one visualization to summarize these patterns.

2. (Exploratory Analysis) How does salary differ across major categories such as: job titles or job levels**,** industries**,** education levels. Visualize these differences and interpret whether they seem meaningful from a business/HR perspective.

3. (Focused Exploration) What is the relationship between years of experience and salary? (e.g., linear, diminishing returns, large jumps at certain career stages) Interpret the pattern and discuss possible reasons from an HR or labor economics perspective.

4. (Regression Analysis) Build a multiple linear regression model predicting salary/income using 2 – 4 relevant predictors (e.g., experience, education, industry, company size). Interpret each coefficient clearly in business terms, explaining how each factor influences salary while holding others constant.

Dataset 3: Customer Personality Analysis

This dataset contains customer demographic information, lifestyle attributes, and detailed spending in various product categories (wine, meat, fruit, gold products, sweets, etc.). It is commonly used to analyze customer segmentation, spending behavior, and campaign effectiveness.

The business objective is to understand which customer characteristics drive spending and how marketers can better target campaigns.

You can download this dataset in this Kaggle page, which provides a detailed data dictionary, variable descriptions under the **Data Card** tab, and hundreds of shared example notebooks under the **Code** tab, which you may use for inspiration (*do not*

copy).

Required Guiding Questions (Answer all)

1. (Descriptive Analysis) What are the key patterns in customer demographics and lifestyle attributes (e.g., income, age, family size, education, recency, online vs. in-store purchases)? Provide descriptive statistics and at least one visualization to summarize overall customer characteristics.

2. (Exploratory Analysis) Do spending patterns (total spending or spending by category) differ across major groups such as: marital status**,** education level**,** number of children**,** campaign acceptance**.** Visualize these differences and interpret whether they are meaningful from a marketing perspective.

3. (Focused Exploration) What is the relationship between household income and total spending? (e.g., linear, plateauing, or outliers such as low-income/high-spend customers)**.** Interpret the pattern and discuss the possible marketing implications.

4. (Regression Analysis) Create a total spending variable (sum of all product-category spending) or choose one major category (e.g., wine spending) as your continuous target for regression. Build a multiple linear regression model predicting spending using 2 – 4 relevant predictors (e.g., income, recency, children, education). Interpret the coefficients clearly in business terms, explaining how each factor influences customer spending while holding other variables constant.

4. Grading Criteria (25% of Final Grade) (Tentative)

1) Data Cleaning & Preparation — 5 points
   - Handles duplicates, missing values, and outliers appropriately
   - Corrects data types and creates useful new variables if needed
   - Clear explanations in markdown

2) Descriptive Analysis — 5 points
   - Summarizes key variables using appropriate statistical measures
   - Descriptive insights are accurate and connected to the dataset context
   - Not excessive, focused on relevant variables

3) Data Visualization — 5 points
   - Includes the required plots (histogram, boxplot, bar/line, scatter)
   - Plots are clean, labeled, and readable
   - Each visualization includes a brief, meaningful interpretation

4) Regression Analysis — 5 points

- Builds a sensible multiple linear regression model using relevant predictors
- Provides correct interpretation of coefficients and R^2
- Explains results in clear business terms

5) Business Insights & Presentation Quality — 5 points

- Answers all required guiding questions thoroughly
- Provides logical, actionable business insights
- Notebook is well organized and presentation-ready
- Clear delivery during the Week 15 presentation

5. Useful Resource

5.1. Reference Notebooks (Kaggle)

You may learn from publicly shared examples analyzing similar datasets. However, **copying full notebooks is not acceptable**. You must interpret results in your own words and produce your own notebook.

5.2. Jupyter Notebook Examples

- Bike Sharing: Multiple Linear Regression
- EDA To Prediction(DieTanic)

5.3. Course Expectation Reminder

Only techniques covered in class will be credited, including:

- Data cleaning (duplicate, missing value, outlier)
- Descriptive statistics
- Data visualization (hist, boxplot, scatter, line)
- Simple and multiple linear regression

5.4. AI Use Disclosure

If you use AI tools:

- You must disclose **how** and **what** you used at the end
- You remain responsible for verifying correctness

5.5 Need Help?

Visit my office hours (CBPM B-214, Mon & Wed: 4 – 6 PM, Fri: 4 – 5 PM) or email (dchen@wku.edu.cn or dchen@kean.edu.cn) with questions.

[This group project instruction document was prepared with assistance from ChatGPT5.1]